

Jeffery Liu

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Third-year Mechatronics Engineering student seeking a summer engineering internship. Real design, build and test experience beyond coursework: an R&D internship, a licensed vehicle dealer, and a full turbo engine rebuild. Strongest in practical problem solving and mechanical design, and building electronics and controls through UC Motorsport.

EDUCATION

University of Canterbury, BE(Hons) Mechatronics Engineering 2024 - present

- Third year, expected graduation 2027. Second-year grade average: A-.
- Offered two competitive UC entrance scholarships (each around 10 awarded nationally); not taken up after gaining residency.
- Relevant coursework: power electronics, computational mechanical analysis, solid mechanics (stress, strain and deformation), dynamics, engineering design and CAD, electrical systems, control and vibrations (PID, root locus, Bode), embedded systems (Arduino, STM32, C), and fluid mechanics.
- NZ Workplace First Aid Certificate (2026).

EXPERIENCE

Mechatronics Engineering Intern, Holmes Solutions Nov 2025 - Feb 2026

Roadside & Marine sector, R&D consultancy (453.5 hours)

- Owned the D/d ratio investigation end to end, from research through senior sign-off, delivering an interchangeable sleeve upgrade to the redirection bollard that gave reliable test runs and data toward safer mooring guidance.
- Mapped apparatus constraints and verified the design with hand calculations and SolidWorks FEA (senior engineer sign-off); flagged a non-conservative capacity check on the revised load case and escalated it for structural review.
- Produced construction-ready drawings and a staged build plan, and prepared weld joints on site; also supported a second intern's crash-test bogie build and roadside crash-test setup.
- Design was built and passed full-scale testing in 2026; co-authored the resulting technical paper, to be presented at an international marine conference in August 2026.

Reference: Ray Keane, Senior Engineer, ray.keane@holmessolutions.com

Founder & Licensed Vehicle Dealer, Jeff's Auto Retail 2022 - 2026

- Sourced, diagnosed, repaired and sold used vehicles end to end, and prepared vehicles to WOF standard.
- Ran all operations: pricing, listings, customer relations, bookkeeping, tax and compliance. Winding down in 2026.

Performance ECU tuning support, Custom Culture & Tuning 2025 - present

- Support an Auckland-based tuner remotely: datalogging, diagnostics, and in-session mechanical adjustments.

Bicycle Technician & Maintenance Tutor, Adventure Cycles 2019 - 2023

- First and longest-running of several bike workshop roles; diagnosed and repaired drivetrain, brake and suspension systems, and ran monthly public maintenance workshops.

PROJECTS

Turbocharged Engine Rebuild & Swap, 2006 Subaru Legacy GT 2025 - present

- Diagnosed an internal engine failure from running symptoms and a compression test, then chose to build a stronger engine rather than repair the original.
- Spec'd and rebuilt the engine for reliability under boost: upgraded, machined block and heads, stronger bearings, rings, head studs and valve springs, and upgraded cooling and oil systems for sustained load.
- Validated the build through datalogging, running primary monitoring and fault-checking myself and working with an experienced tuner to refine the calibration from logged data; now daily driven, with motorsport-style tune/calibration (anti-lag, flat-foot shifting) planned.

Line Follower Robot, ENMT221 Mechatronics Design 2025

- Designed and 3D-printed the complete mechanical platform and drivetrain in SolidWorks for a robot built from scratch by the team (custom PCB, sensors and motor control), and took part in the on-track testing and tuning.
- Set a practice lap quick enough to have won that year's event, but a motor failure late on left no time to re-tune the replacement, and the team finished 3rd.

Tyre Temperature Sensor, UC Motorsport (FSAE) 2026 - present

- Junior team member developing an onboard tyre-temperature logging system, intended to feed a variable friction coefficient into the car's new traction control system. Schematic, PCB layout and BOM complete, DFM-passed, hardware on order.

RoboCup Autonomous Robot, ENMT301 Mechatronics System Design 2026 - present

- Team of three building a fully autonomous robot for a timed arena challenge: navigating and avoiding obstacles, collecting target weights, and returning them to base.
- Working across mechanical design, CAD and electrical, with autonomous navigation a key focus; proposed the design concept the team chose to build. Build in progress: ramp and drum intake through multiple iterations, ToF sensing linked to drive control for obstacle response.

SKILLS

Mechanical & practical: Diagnostics & repair, engine building, MIG & arc welding, fabrication, SolidWorks (incl. FEA), 3D printing

Electronics & hardware: Soldering, auto & marine wiring, PCB design (KiCad)

Controls & embedded (working knowledge): PLC (ladder logic), embedded architecture, control theory, ECU tuning support, datalogging

Business & communication: Small business management, sales, bilingual (English/Mandarin)